TECHNICAL DILIGENCE REVIEW - RECORD 015

Helioscope Robotics, Inc. - diligence file reference TECHNICAL_DILIGENCE-015

1. Scope of the component and licence review

Brandywine Technology Partners was instructed by Ardent Peak to examine the third-party component inventory, licence exposure and contribution provenance of the codebase; the resulting report is the report at TDR-2026-07, dated 2026-03-17. Report TDR-2026-07 was prepared by Brandywine Technology Partners for Ardent Peak and covers the third-party component inventory, licence exposure and contribution provenance of the codebase at Helioscope Robotics, Inc. as at 2026-03-17. The reviewer asked in each case whether a declined item had a compensating control, and the present report reports the answer even where it is unsatisfactory. Management's declinatures are reported in report TDR-2026-07 at face value; the reviewer's disagreement is stated separately and does not overwrite them. TDR-2026-07 deliberately preserves detail that a summary would drop, because the detail is where the qualifications live.

Report TDR-2026-07 added two components to its sample after the first week, because the interview evidence pointed at them. The present report covers 15 of 34 services; the untested remainder is listed so a reader can see what was not reached. 2026-03-17 is the evidence cut-off for TDR-2026-07, whose subject is the third-party component inventory, licence exposure and contribution provenance of the codebase. The subject of the present report is the third-party component inventory, licence exposure and contribution provenance of the codebase, examined for Ardent Peak by Brandywine Technology Partners over 4 weeks ending 2026-03-17. The third-party component inventory, licence exposure and contribution provenance of the codebase produced more evidence than this report reproduces, and the working papers hold the remainder.

TDR-2026-07 follows the reviewer's usual practice of grading evidence strength separately from finding severity, so a plausible concern is not promoted to a verified defect. M. Quillan was granted read access to the code, the pipeline and the incident tracker for the purposes of this report; production access was not requested. 10 matters remain open at the date of the present report, and none of them is closed by the passage of time alone. M. Quillan has dated each open item in report TDR-2026-07 so that a reader can see which have been open longest rather than only how many there are. M. Quillan has kept the working papers behind this report in a form that a second reviewer could re-perform.

The present report does not include a code audit; the reviewer read code where a specific claim required it and no further. Nothing in this report rests on a test the reviewer designed; the tests were the company's, and the reviewer read the results and the method. The working method behind TDR-2026-07 was to start from the incident record and work back to the design, rather than the other way round. The present report did not rely on management representation alone; every material claim was traced to a document, a measurement or an observation. Brandywine Technology Partners raised each finding with the owning engineer before it was written up, and the responses are recorded.

The evidence convention in the present report is that severity describes consequence and evidence strength describes confidence, and the two are never merged. M. Quillan has left several concerns in report TDR-2026-07 at limited evidence strength rather than dropping them or dressing them up. In report TDR-2026-07 the reviewer distinguishes what was verified from what was represented, and marks each conclusion accordingly. Work on report TDR-2026-07 ran over 4 weeks and included 11 interviews and read-only access to the engineering repositories. The evidence behind TDR-2026-07 is retained by Brandywine Technology Partners and is available to Ardent Peak on request.

10 follow-up items arose from the interview programme for the report at TDR-2026-07; 1 were still open at delivery. 8 teams contributed to the interview programme for report TDR-2026-07, and one of them was interviewed twice after a document arrived late. The instruction behind this report was narrow: examine the third-party component inventory, licence exposure and contribution provenance of the codebase and report by 2026-03-17. The present report is one of a series of technical reports; this one takes the third-party component inventory, licence exposure and contribution provenance of the codebase and reports against the evidence available by 2026-03-17. The third-party component inventory, licence exposure and contribution provenance of the codebase is reported here at the level of detail Ardent Peak asked for and no deeper.

Schedule T - Third-party component and licence inventory (TDR-2026-07)

Component	Version	Licence	Copyleft	Usage	Approval	Consuming service
nav-graph-solver	2.4.1	Apache-2.0	no	linked into the planner as a static library	approved 2024-02-19	Motion Planner
geom2d-fusion	0.9.7	LGPL-2.1	weak	dynamically linked by the fusion pipeline	approved 2023-08-04	Costmap Fusion
scanmatch-legacy	1.2.0	GPL-2.0	strong	linked into the localization fallback path	not approved	Localization Engine
grpc-transport	1.62.0	Apache-2.0	no	service-to-service transport across the cloud	approved 2024-05-30	Event Bus
tsdb-client	3.1.4	MIT	no	telemetry writes from the ingest service	approved 2024-09-11	Telemetry Ingest
chartkit	4.8.2	BSD-3-Clause	no	operator console visualisations	approved 2023-11-27	Remote Operations Console
physics-core-fork	0.14.3	Zlib	no	simulation physics	approved with conditions 2023-04-18	Simulation Harness
rtos-shim	1.0.9	BSD-2-Clause	no	real-time helper layer on the robot	approved 2024-01-23	Motion Planner
protobuf-runtime	25.3	BSD-3-Clause	no	message encoding across edge and cloud	approved 2024-04-08	Config Distribution
sqlite-embedded	3.45.1	public domain dedication	no	on-robot local state	approved 2023-06-15	Edge Gateway Agent
opencv-subset	4.9.0	Apache-2.0	no	image preprocessing for grasp proposals	approved 2024-07-02	Gripper Controller
pointcloud-utils	2.1.6	MPL-2.0	weak	point cloud filtering in fusion	approved 2024-03-14	Costmap Fusion
yaml-parse	0.15.2	MIT	no	configuration parsing on the robot	approved 2023-09-29	Config Distribution
jwt-verify	5.2.1	MIT	no	token verification in the identity service	approved 2024-06-21	Identity and Authorization
kafka-client	3.6.1	Apache-2.0	no	event bus producer and consumer clients	approved 2024-02-27	Event Bus
media-transcode	6.0.0	LGPL-2.1 with an optional GPL build	weak	video snippet transcoding	approved with conditions 2024-08-16	Video Snippet Service
wms-adapter-sdk	2.2.0	commercial, per-seat	no	one customer's warehouse system adapter	approved 2023-12-05	WMS Connector
lidar-vendor-sdk	4.1.2	commercial, redistributable binary	no	lidar driver	approved 2023-05-22	Lidar Driver
safety-controller-toolchain	3.0.4	commercial, development use only	no	safety controller build toolchain	approved 2023-02-10	Safety Controller
cell-model-lib	1.8.0	Apache-2.0	no	battery state estimation	approved 2024-10-01	Battery Management Service
grid-render	3.3.5	GPL-3.0 with a commercial alternative	strong	map authoring geometry rendering	approved with conditions 2023-07-19	Map Authoring Studio
http-router	1.9.4	MIT	no	cloud service routing	approved 2024-01-09	Order Broker
otel-instrument	1.24.0	Apache-2.0	no	tracing across cloud services	approved 2024-05-14	Fleet Health Monitor
pgbouncer-managed	1.21.0	ISC	no	database connection pooling	approved 2024-03-05	Fleet Allocator
wheel-slip-model	0.3.1	AGPL-3.0	strong	prototype slip	not approved	Wheel Odometry

Component	Version	Licence	Copyleft	Usage	Approval	Consuming service
				detector, not shipped		Filter
csv-export	2.0.3	MIT	no	billing and usage exports	approved 2023-10-17	Billing Usage Meter
pager-provider-sdk	3.4.0	commercial, subscription	no	alarm dispatch	approved 2023-03-28	Alarm Router
k8s-operator-framework	1.31.0	Apache-2.0	no	cloud deployment controllers	approved 2024-09-24	Site Provisioning Service
image-signing	2.5.0	Apache-2.0	no	artefact signature verification	approved 2025-01-15	OTA Updater
warehouse-sim-assets	1.1.0	CC BY-SA 4.0	strong	simulation scenario assets	approved with conditions 2024-11-06	Simulation Harness

2. Component inventory and licence classification

Protobuf-runtime carries this qualification: attribution present in both notice files. Compliance with the protobuf-runtime terms requires to reproduce the notice text in the delivered documentation. Protobuf-runtime also carries a fork that must be maintained locally for as long as it is used, which is a supportability question rather than a licence one. protobuf-runtime was tested rather than accepted, and the test is described in the working papers behind TDR-2026-07.

Version 3.0.4 of safety-controller-toolchain is the catalogued build at OSS-019, used safety controller build toolchain. Compliance with the safety-controller-toolchain terms requires nothing beyond retaining the licence text with the source. Safety-controller-toolchain affects Safety Controller; no other component reviewed depends on it. safety-controller-toolchain was cross-checked against the engineering record supplied for TDR-2026-07, and the two agree.

The operational obligation attaching to pager-provider-sdk is to keep the dynamic-linking boundary intact. Pager-provider-sdk shows an approval status of approved 2023-03-28 against the company's own policy. Against pager-provider-sdk the reviewer would add that the older application interface is deprecated by the provider with an end date in 2027. pager-provider-sdk was tested rather than accepted, and the test is described in the working papers behind TDR-2026-07.

Cell-model-lib is recorded as approved 2024-10-01 in the component approval log. Cell-model-lib was obtained as follows: public registry, signature verified. A commercial supplier whose support statement expires inside the plan period attaches to cell-model-lib independently of its licence terms. cell-model-lib is reported at the evidence strength available and has not been promoted to fit the narrative.

Linking mode matters for wheel-slip-model, which is used prototype slip detector, not shipped. Where wheel-slip-model came from matters here: public registry. Wheel-slip-model carries this qualification: a network-copyleft component in a prototype; the reviewer flagged the shipping risk before validation completes. wheel-slip-model is one of the entries where TDR-2026-07 relies on the company's own measurement rather than an independent one.

Kafka-client carries Apache-2.0 terms, classified no for copyleft purposes. Kafka-client presents a build-time dependency whose absence would stop the pipeline rather than the product to the engineering team. The reviewer's note on kafka-client is that one service still vendors a 6.x client. kafka-client has no single accountable owner, and that is the finding the reviewer would press hardest.

For pointcloud-utils the obligation is to keep the dynamic-linking boundary intact, and the reviewer tested whether it is met. Beyond licensing, pointcloud-utils carries an upstream project with no active maintainer. Pointcloud-utils carries the approval entry approved 2024-03-14; the reviewer looked for a corresponding decision record. pointcloud-utils was inspected in the repository rather than described from the ticket, which is why the account is specific.

Against scanmatch-legacy the reviewer would add that strong-copyleft exposure in shipped robot software; the reviewer treats this as the principal IP finding. Scanmatch-legacy was assessed for support as well as licence risk, and it carries a single-maintainer dependency with no succession plan. Scanmatch-legacy shows an approval status of not approved against the company's own policy. scanmatch-legacy was checked against the company's own register, which had not been updated since the change landed.

Compliance with the grpc-transport terms requires to share alike any derived asset it distributes. The non-licence risk on grpc-transport is a component with open advisories that do not affect the deployed configuration. Grpc-transport

carries the approval entry approved 2024-05-30; the reviewer looked for a corresponding decision record. grpc-transport appears twice in the working papers for TDR-2026-07 because the first answer was corrected after a follow-up request.

In addition to its terms, chartkit brings an upstream project that is active but changes interface frequently. For chartkit the obligation is to publish the modified files it distributes, and the reviewer tested whether it is met. On provenance, chartkit shows public registry; upstream archived in 2025. chartkit is reported here in full because a summary of it in an earlier draft of TDR-2026-07 was misread as an assurance.

Schedule C - Producer and consumer interface contracts examined (TDR-2026-07)

Producer service	Consumer	Interface form	Interface version	Compatibility check	Breaking changes
Motion Planner	traffic manager	internal protocol	v2.8	enforced in the pipeline	2 in the reviewed period
Motion Planner	on-robot drive supervisor	request-response interface	v2.7	enforced in the pipeline	2 in the reviewed period
Costmap Fusion	motion planner	published schema	v3.9	not automated	1 in the reviewed period
Costmap Fusion	speed	streamed topic	v2.10	enforced in the pipeline	none
Fleet Allocator	traffic manager	request-response interface	v4.10	not automated	1 in the reviewed period
Fleet Allocator	fleet health monitor	shared event contract	v1.7	enforced in the pipeline	1 in the reviewed period
Traffic Manager	motion planner	shared event contract	v3.4	advisory only	2 in the reviewed period
Traffic Manager	remote operations console	streamed topic	v3.4	enforced in the pipeline	none
Charge Scheduler	fleet allocator	published schema	v2.6	not automated	none
Charge Scheduler	fleet health monitor	request-response interface	v1.8	not automated	2 in the reviewed period
Safety Controller	drive power contactors	published schema	v2.5	advisory only	1 in the reviewed period
Safety Controller	robot fault annunciator	streamed topic	v1.2	enforced on the producer side	1 in the reviewed period
E-Stop Chain	safety controller	internal protocol	v4.9	not automated	2 in the reviewed period
E-Stop Chain	drive contactors	shared event contract	v4.5	not automated	1 in the reviewed period
Lidar Driver	costmap fusion pipeline	streamed topic	v4.8	advisory only	none
Lidar Driver	localization engine	published schema	v3.1	advisory only	2 in the reviewed period
Localization Engine	motion planner	streamed topic	v2.0	enforced on the producer side	1 in the reviewed period
Localization Engine	fleet health monitor	published schema	v2.7	not automated	1 in the reviewed period
WMS Connector	order broker	streamed topic	v4.1	not automated	none
WMS Connector	billing usage meter	shared event contract	v1.2	not automated	none
Order Broker	fleet allocator	request-response interface	v1.4	enforced in the pipeline	2 in the reviewed period
Order Broker	analytics warehouse loader	published schema	v4.11	enforced on the producer side	1 in the reviewed period
Telemetry Ingest	fleet health monitor	internal protocol	v3.8	enforced in the pipeline	none
Telemetry Ingest	alarm router	request-response interface	v2.7	enforced in the pipeline	1 in the reviewed period
OTA Updater	on-robot supervisor	shared event contract	v1.7	not automated	2 in the reviewed period
OTA Updater	fleet health monitor	streamed topic	v2.10	enforced on the producer side	2 in the reviewed period
Identity and Authorization	every cloud service	shared event contract	v4.10	enforced on the producer side	2 in the reviewed period
Identity and Authorization	remote operations console	published schema	v1.8	advisory only	1 in the reviewed period
Remote Operations Console	traffic manager	request-response interface	v2.11	not automated	none
Remote Operations Console	alarm router	published schema	v3.0	enforced on the producer side	none
Simulation Harness	release orchestrator gate	request-response interface	v4.10	enforced on the producer side	2 in the reviewed period

Producer service	Consumer	Interface form	Interface version	Compatibility check	Breaking changes
Simulation Harness	autonomy regression report	request-response interface	v2.4	not automated	2 in the reviewed period
Hardware-in-the-Loop Rig	firmware release decision record	published schema	v2.3	enforced on the producer side	none
Fleet Health Monitor	alarm router	streamed topic	v4.10	enforced on the producer side	none
Fleet Health Monitor	field service workflow	published schema	v4.6	advisory only	none
Battery Management Service	charge scheduler	request-response interface	v4.11	not automated	2 in the reviewed period
Battery Management Service	fleet health monitor	streamed topic	v2.1	enforced on the producer side	1 in the reviewed period
Gripper Controller	order execution path	published schema	v2.4	enforced in the pipeline	2 in the reviewed period
Gripper Controller	safety controller fault input	published schema	v3.5	enforced on the producer side	1 in the reviewed period
Map Authoring Studio	map distribution channel	internal protocol	v3.4	advisory only	2 in the reviewed period
Map Authoring Studio	localization engine	internal protocol	v1.8	not automated	1 in the reviewed period
Site Provisioning Service	config distribution service	published schema	v1.6	enforced on the producer side	none
Site Provisioning Service	WMS connector	internal protocol	v2.10	enforced in the pipeline	1 in the reviewed period
Event Bus	every consuming service	published schema	v1.5	advisory only	2 in the reviewed period
Config Distribution	on-robot supervisor	streamed topic	v2.1	enforced on the producer side	1 in the reviewed period
Config Distribution	motion planner parameters	internal protocol	v1.3	enforced on the producer side	1 in the reviewed period
Robot Shadow Store	fleet allocator	shared event contract	v4.9	enforced on the producer side	2 in the reviewed period
Robot Shadow Store	remote operations console	streamed topic	v1.9	enforced on the producer side	1 in the reviewed period
Analytics Warehouse Loader	customer reporting views	streamed topic	v4.2	enforced in the pipeline	none
Analytics Warehouse Loader	internal metrics layer	shared event contract	v2.1	enforced on the producer side	2 in the reviewed period
Billing Usage Meter	finance export	streamed topic	v2.7	enforced in the pipeline	1 in the reviewed period
Billing Usage Meter	customer usage report	streamed topic	v3.8	advisory only	none
Alarm Router	paging provider	streamed topic	v2.3	not automated	2 in the reviewed period
Alarm Router	customer notification channel	request-response interface	v3.6	not automated	none
Video Snippet Service	remote operations console	shared event contract	v2.5	enforced in the pipeline	none
Video Snippet Service	incident review workflow	shared event contract	v3.9	enforced in the pipeline	1 in the reviewed period
Speed and Separation Monitor	safety controller	streamed topic	v1.3	enforced in the pipeline	1 in the reviewed period
Speed and Separation Monitor	motion planner speed limit	published schema	v1.6	advisory only	1 in the reviewed period
Wheel Odometry Filter	localization engine	request-response interface	v1.2	not automated	1 in the reviewed period
Wheel Odometry Filter	motion planner	published schema	v2.7	enforced on the producer side	2 in the reviewed period
Payload Scale Interface	gripper controller	request-response interface	v2.0	enforced on the producer side	2 in the reviewed period
Payload Scale Interface	order execution path	shared event contract	v4.9	not automated	1 in the reviewed period
Edge Gateway Agent	telemetry ingest path	streamed topic	v4.3	enforced in the pipeline	2 in the reviewed period
Edge Gateway Agent	robot shadow store	streamed topic	v2.1	advisory only	1 in the reviewed period
Release Orchestrator	OTA updater	request-response interface	v1.4	enforced in the pipeline	none
Release Orchestrator	cloud deployment controllers	internal protocol	v4.0	advisory only	1 in the reviewed period

3. Copyleft exposure and the shipped-software boundary

At OSS-010 the component is sqlite-embedded, pinned at 3.45.1 and used on-robot local state. Under public domain dedication, sqlite-embedded is treated as no copyleft in the inventory. Compliance with the sqlite-embedded terms requires nothing beyond retaining the licence text with the source. sqlite-embedded is reported at the evidence strength available and has not been promoted to fit the narrative.

A build-time dependency whose absence would stop the pipeline rather than the product attaches to jwt-verify independently of its licence terms. Public registry, signature verified is the provenance recorded for jwt-verify. Jwt-verify carries a practical obligation: to reproduce the notice text in the delivered documentation. jwt-verify has no single accountable owner, and that is the finding the reviewer would press hardest.

Public registry, signature verified is the provenance recorded for csv-export. Csv-export version 2.0.3 is catalogued at OSS-026 and is used billing and usage exports. Attribution present is the reviewer's observation on csv-export. csv-export would move from a concern to a verified defect if one further measurement were supplied.

The open question on image-signing is that attribution present. Image-signing also carries no identified risk beyond ordinary version currency, which is a supportability question rather than a licence one. In the approval log, image-signing is approved 2025-01-15. image-signing was the subject of a specific management representation, which the reviewer has recorded but not relied on alone.

To attribute the component in the customer-facing notice file is what opencv-subset requires in practice. Public registry, signature verified is the provenance recorded for opencv-subset. Licence text and the actual usage of opencv-subset were read together, and notice reproduced in the robot documentation. opencv-subset was one of two items where TDR-2026-07 could not obtain the underlying data within the engagement.

On provenance, physics-core-fork shows forked from upstream in 2023 with local patches. The distribution question for physics-core-fork turns on the fact that it is used simulation physics. In the approval log, physics-core-fork is approved with conditions 2023-04-18. physics-core-fork sits inside the sample rather than being an exception to it, which is the reason it is reported at length.

Http-router sits inside Order Broker and is therefore in the shipped product rather than in tooling. Usage of http-router 1.9.4 is cloud service routing, and the inventory reference is OSS-022. Licence text and the actual usage of http-router were read together, and attribution present. http-router illustrates why TDR-2026-07 separates the severity of a finding from the confidence behind it.

Compliance with the tsdb-client terms requires to keep the dynamic-linking boundary intact. Tsdb-client version 3.1.4 is catalogued at OSS-005 and is used telemetry writes from the ingest service. Approved 2024-09-11 is the approval entry for tsdb-client. tsdb-client rests on a single source, and the reviewer has said so rather than implying corroboration.

Licence text and the actual usage of warehouse-sim-assets were read together, and share-alike terms apply to derived assets; four internally modified assets are not published. Warehouse-sim-assets is recorded as approved with conditions 2024-11-06 in the component approval log. In Simulation Harness the component warehouse-sim-assets is used simulation scenario assets, which is the fact the licence analysis rests on. warehouse-sim-assets was tested rather than accepted, and the test is described in the working papers behind TDR-2026-07.

OSS-001 covers nav-graph-solver at version 2.4.1, used linked into the planner as a static library. Provenance of nav-graph-solver is public registry, signature verified. Policy compliance for nav-graph-solver rests on its approval entry, which is approved 2024-02-19. nav-graph-solver was compared with the equivalent position at two other components, and it is neither the best nor the worst.

Schedule L - Vulnerability ageing against the internal remediation window (TDR-2026-07)

Finding	Severity	Days open at review	Target window	Within target	Compensating control
SEC-2025-01	High	22	30 days	yes	none recorded
SEC-2025-02	High	142	30 days	no	manual review before release
SEC-2025-03	Medium	87	90 days	yes	monitoring alert on use
SEC-2025-04	Medium	43	90 days	yes	monitoring alert on

Finding	Severity	Days open at review	Target window	Within target	Compensating control
					use
SEC-2025-05	High	76	30 days	no	none recorded
SEC-2025-06	Medium	22	90 days	yes	second delivery path
SEC-2025-07	Medium	24	90 days	yes	second delivery path
SEC-2025-08	Medium	154	90 days	no	network segmentation
SEC-2025-09	Medium	308	90 days	no	second delivery path
SEC-2025-10	Low	167	180 days	yes	audit trail only
SEC-2025-11	Low	78	180 days	yes	restricted role membership
SEC-2025-12	Low	354	180 days	no	network segmentation
SEC-2026-01	High	232	30 days	no	monitoring alert on use
SEC-2026-02	High	127	30 days	no	second delivery path
SEC-2026-03	High	27	30 days	yes	monitoring alert on use
SEC-2026-04	Medium	135	90 days	no	second delivery path
SEC-2026-05	Medium	100	90 days	no	second delivery path
SEC-2026-06	Medium	192	90 days	no	monitoring alert on use
SEC-2026-07	Medium	158	90 days	no	second delivery path
SEC-2026-08	Medium	151	90 days	no	none recorded
SEC-2026-09	Low	150	180 days	yes	second delivery path
SEC-2026-10	Low	78	180 days	yes	network segmentation
SEC-2026-11	Low	200	180 days	no	manual review before release
SEC-2026-12	Low	252	180 days	no	second delivery path
SEC-2026-13	Low	24	180 days	yes	second delivery path
SEC-2026-14	Low	128	180 days	yes	restricted role membership
SEC-2026-15	Low	127	180 days	no	manual review before release
SEC-2026-16	Medium	148	90 days	no	manual review before release
SEC-2026-17	High	147	30 days	no	none recorded
SEC-2026-18	Low	305	180 days	no	monitoring alert on use
SEC-2026-19	Medium	116	90 days	no	audit trail only
SEC-2026-20	Low	97	180 days	no	none recorded

4. Where third-party code sits inside the product

Test coverage on Identity and Authorization is 84 per cent line coverage and the measurement basis was confirmed with Sonia Krall. Identity and Authorization runs cloud, global control plane with regional caches; that placement is what determines whether a fault is one robot or one region. Identity and Authorization was assessed against the objective and the defect record, and the design is sound and the test evidence is thinner than the design deserves. The Identity and Authorization codebase is Go 1.22 throughout, deployed onto cloud, containers on a managed Kubernetes cluster, and its durable record is PostgreSQL 14 with a hardware-backed key store. The customer directory federations and the secret store feed Identity and Authorization; a failure in Identity and Authorization is felt by every cloud service and the remote operations console. Identity and Authorization was inspected in the repository rather than described from the ticket, which is why the account is specific.

The evidence base for Identity and Authorization was the authorization model document, a token lifetime audit and the secret rotation log. Identity and Authorization carries 6 open defects, among them a case where service-to-service tokens are issued with a twelve-hour lifetime, longer than the design intent. Identity and Authorization was checked for shared-database coupling, and the ownership boundary held. Security Engineering releases Identity and Authorization weekly, with a security review on every change, and no release in the sampled window bypassed the gate set. Security Engineering holds Identity and Authorization, and the ownership entry naming Sonia Krall matches the on-call rotation. Identity and Authorization was the subject of a specific management representation, which the reviewer has recorded but not relied on alone.

Upstream of Traffic Manager sit the fleet allocator and the site map service; downstream, the motion planner and the remote operations console consume its output. Traffic Manager is owned by the Fleet Services team, with Owen Brantley named as the engineering owner of record. The open defect backlog for Traffic Manager holds 12 items, the most consequential being that a corridor reservation leaks when a robot drops off the network mid-traverse. Hosting for Traffic Manager is cloud, containers on a managed Kubernetes cluster, and its Go 1.22 code writes through to Redis 7 for corridor reservations with a PostgreSQL 14 audit trail. Traffic Manager is promoted weekly, paired with the allocator release, with the ring order fixed by policy. Traffic Manager appears twice in the working papers for TDR-2026-07 because the first answer was corrected after a follow-up request.

Test coverage on Traffic Manager is 74 per cent line coverage and the measurement basis was confirmed with Owen Brantley. The assessment of Traffic Manager is not unqualified: the position is stronger than the coverage figure alone would suggest, because the bench evidence is unusually good here. The published objective for Traffic Manager is that corridor deadlock detected and broken inside two seconds. Traffic Manager runs cloud, regional, with a degraded on-robot yield behaviour if the link drops; that placement is what determines whether a fault is one robot or one region. Evidence examined for Traffic Manager comprised the reservation protocol specification and a replay of three deadlock events. Traffic Manager has an owner who accepted the finding immediately, which is worth recording alongside the finding itself.

Safety Controller's dependency ageing was checked: none; the supplier's support commitment runs to 2031. Taking the evidence together, Safety Controller the component would not block the transaction, and it would appear on any competent post-close engineering plan. Safety Engineering tracks 3 open defects on Safety Controller, including the acknowledged case where the fault log wraps after four thousand entries and the oldest records are discarded. Safety Controller runs edge, dual-channel, with cross-checking between channels; that placement is what determines whether a fault is one robot or one region. Safety Controller sits at 91 per cent line coverage; the trend across the reviewed period was up by three points. Safety Controller is the clearest instance in TDR-2026-07 of a control that exists on paper and is weaker in practice.

Safety Engineering releases Safety Controller twice a year, with full revalidation of the safety case, and no release in the sampled window bypassed the gate set. Ownership of Safety Controller sits with Safety Engineering; Bettina Vogel signs off changes to it. Safety Controller rests on the supplier assessment report, the internal safety case and eight bench trigger tests so far as this review is concerned. The dependency map places the E-stop chain, the speed and separation monitor and the drive current sensors above Safety Controller and the drive power contactors and the robot fault annunciator below it. Safety Controller does not share a schema with any consumer; every read by the drive power contactors and the robot fault annunciator goes through the published interface. Safety Controller carries a follow-up request that was satisfied after the draft of TDR-2026-07 was circulated.

The Analytics Warehouse Loader codebase is Python 3.11 with SQL transformations throughout, deployed onto cloud, a scheduled batch runner, and its durable record is a columnar warehouse with daily partitions. The interface Analytics Warehouse Loader publishes is versioned, and the customer reporting views and the internal metrics layer were checked for use of any undocumented field. Data and Telemetry publishes one objective for Analytics Warehouse Loader: the daily load completes before 06:00 local time at every region. Against Analytics Warehouse Loader, the coverage figure is 45 per cent line coverage, which the reviewer notes that the untested share includes the error handling. In production Analytics Warehouse Loader is cloud, one loader per region writing to a shared warehouse, unchanged since the last architecture review. Analytics Warehouse Loader was compared with the equivalent position at two other components, and it is neither the best nor the worst.

Underlying Analytics Warehouse Loader in this report are the load schedule, partition completeness checks and two months of load audit records. The backlog for Analytics Warehouse Loader is 20 items deep and contains an acknowledged defect whereby a late-arriving telemetry partition is silently excluded rather than reprocessed. The telemetry ingest path, the order broker and the billing usage meter feed Analytics Warehouse Loader; a failure in Analytics Warehouse Loader is felt by the customer reporting views and the internal metrics layer. Responsibility for Analytics Warehouse Loader was confirmed with Farida Cisse, whose team, Data and Telemetry, also holds the on-call duty. Supportability of Analytics Warehouse Loader turns on one fact: one transformation still runs on a deprecated

warehouse function. Analytics Warehouse Loader is not material on its own and belongs in this report because of what it reveals about the surrounding process.

The cadence recorded for Robot Shadow Store is weekly, which matches the observed release history. The Fleet Services team carries Robot Shadow Store on its service inventory, and Marta Kellerman answered for it in interview. Coverage for Robot Shadow Store was 68 per cent line coverage at the review date, and the trend over the reviewed period was up by seven points. For Robot Shadow Store the supportability position is none identified. Data ownership is clear for Robot Shadow Store: it writes Redis 7 as the hot path with PostgreSQL 14 as the durable record and no other service writes there. Robot Shadow Store is not material on its own and belongs in this report because of what it reveals about the surrounding process.

Robot Shadow Store was built in Go 1.22 and has not been rewritten; it still runs on cloud, containers with a sharded cache against Redis 7 as the hot path with PostgreSQL 14 as the durable record. Robot Shadow Store sits cloud, regional, sharded by site, and no alternative placement is currently supported. Evidence examined for Robot Shadow Store comprised the shadow consistency note, staleness measurements and a rebalance observation. Robot Shadow Store was assessed against the objective and the defect record, and the reviewer could corroborate the availability claim but not the latency claim, which rests on a single dashboard. The fleet allocator, the traffic manager and the remote operations console rely on Robot Shadow Store, which in turn relies on the telemetry ingest path and the edge gateway agent. Robot Shadow Store is the clearest instance in TDR-2026-07 of engineering practice being better than the documentation suggests.

Interface discipline around Localization Engine is adequate; the schema is versioned and change is gated on a compatibility check. Deployment of Localization Engine is edge, with map distribution from the cloud, which sets the blast radius of any defect in it. Localization Engine pairs a C++17 with a Rust bridge layer implementation with on-robot Linux, pinned to a dedicated core, and its data lives in a local map cache with a periodic checkpoint. Localization Engine rests on the localization design note, survey reports from three sites and the reviewer's map audit so far as this review is concerned. Test coverage on Localization Engine is 66 per cent line coverage and the measurement basis was confirmed with Rahul Menon. Localization Engine is included because a reader of TDR-2026-07 should be able to disagree with the inference without repeating the work.

Taking the evidence together, Localization Engine the position is stronger than the coverage figure alone would suggest, because the bench evidence is unusually good here. Localization Engine sits between the lidar driver, the wheel odometry filter and the map distribution channel and the motion planner and the fleet health monitor, and that position is what makes its availability objective load-bearing. The objective recorded against Localization Engine is that pose error stays under 25 mm in mapped aisles at survey speed, and it is reported weekly to the engineering leadership. Ownership of Localization Engine sits with Autonomy Core; Rahul Menon signs off changes to it. 16 defects remain open against Localization Engine; the one the reviewer pressed on is that relocalisation after a lift-and-place takes longer than the operator prompt allows. Localization Engine was discussed with two engineers who gave slightly different accounts, and the difference is preserved here.

Map Authoring Studio depends on the site survey imports and the identity service and is depended on by the map distribution channel and the localization engine. What the reviewer had for Map Authoring Studio was the authoring workflow document, the revision history for six sites and an export audit; the conclusions below do not reach past it. Map Authoring Studio carries a single measurable objective, namely that a published map revision reaches every robot at the site inside fifteen minutes. Implementation of Map Authoring Studio rests on TypeScript with a Python service layer; the process runs on cloud, a desktop-class web application and persists to PostgreSQL 14 with map revisions in object storage. Map Authoring Studio reports 49 per cent line coverage, measured by the same instrumentation the pipeline uses. Map Authoring Studio was cross-checked against the engineering record supplied for TDR-2026-07, and the two agree.

22 items sit in the Map Authoring Studio backlog; the entry that bears on the investment case is that an unpublished draft can be exported and hand-loaded, bypassing the review step. One coupling concern applies to Map Authoring Studio: PostgreSQL 14 with map revisions in object storage is shared with a second service, so a schema change is not a local decision. The assessment of Map Authoring Studio is not unqualified: management's own assessment is broadly supported, with the coverage gap treated as a genuine finding rather than a presentational one. Lucia Ferrante leads Applications's work on Map Authoring Studio and was the reviewer's principal interview for it. On end-of-life exposure,

Map Authoring Studio shows a geometry rendering library one major version behind its supported line. Map Authoring Studio sits inside the sample rather than being an exception to it, which is the reason it is reported at length.

Simulation Harness is deployed cloud batch, with a local mode for developers. The release orchestrator gate and the autonomy regression report rely on Simulation Harness, which in turn relies on the planner and localization builds under test. Simulation Harness carries a single measurable objective, namely that the nightly scenario suite completes inside four hours. Hosting for Simulation Harness is developer workstations and a batch cluster, and its Python 3.11 with a C++ physics core code writes through to object storage for scenario definitions and run artefacts. Simulation Harness carries 31 open defects, among them a case where scenario determinism breaks when the physics core is rebuilt with a different optimisation level. Simulation Harness was excluded from an earlier draft of TDR-2026-07 and reinstated after the reviewer obtained the supporting record.

For Simulation Harness the reviewer worked from the scenario catalogue, two weeks of nightly run records and a determinism experiment. On paper Simulation Harness releases continuous, from the main branch, and the observed practice did not diverge from that. Simulation Harness reports 47 per cent line coverage, measured by the same instrumentation the pipeline uses. Responsibility for Simulation Harness was confirmed with Colin Achebe, whose team, Developer Productivity, also holds the on-call duty. Simulation Harness was weighed on its coverage, its defect backlog and its dependency position, and the reviewer could corroborate the availability claim but not the latency claim, which rests on a single dashboard. Simulation Harness was discussed with two engineers who gave slightly different accounts, and the difference is preserved here.

Schedule O - Robot variant, module revision and bench rig coverage (TDR-2026-07)

Robot variant	Module	Firmware revision	Sites deployed	Bench rig coverage	Revision status
HR-240 tote mover	navigation lidar	2.10.0	3	rig one only	one revision behind
HR-300 pallet mover	navigation lidar	3.13.7	7	rig two only	current
HR-320 pallet mover	network radio	4.8.8	14	both rigs	one revision behind
HR-200 tote mover	safety scanner	3.6.6	5	both rigs	current
HR-410 manipulator	depth camera	3.18.5	4	both rigs	one revision behind
HR-300 pallet mover	E-stop harness	1.15.0	14	neither rig	current
HR-240 tote mover	load cell deck	2.14.2	5	neither rig	current
HR-300 pallet mover	network radio	3.16.2	3	rig two only	current
HR-240 tote mover	E-stop harness	4.11.5	4	rig one only	one revision behind
HR-320 pallet mover	load cell deck	2.11.0	8	rig two only	one revision behind
HR-200 tote mover	gripper coprocessor	1.8.2	4	rig one only	pre-production
HR-200 tote mover	charge contact assembly	1.15.7	8	neither rig	current
HR-300 pallet mover	safety scanner	4.8.4	13	both rigs	superseded
HR-300 pallet mover	depth camera	2.16.2	11	rig one only	one revision behind
HR-240 tote mover	charge contact assembly	4.5.1	3	both rigs	superseded
HR-240 tote mover	depth camera	2.17.6	11	neither rig	one revision behind
HR-200 tote mover	charge contact assembly	4.0.3	10	neither rig	pre-production
HR-320 pallet mover	navigation lidar	1.10.1	7	rig one only	one revision behind
HR-240 tote mover	compute module	4.9.2	10	rig one only	current
HR-200 tote mover	network radio	4.16.1	4	rig two only	pre-production
HR-200 tote mover	wheel encoder set	2.10.6	5	both rigs	pre-production
HR-300 pallet mover	load cell deck	1.4.5	6	rig two only	pre-production

4. Where third-party code sits inside the product - part 2 of 2

The 4. Where third-party code sits inside the product begun above continues from this point, and its row references run consecutively across the parts rather than restarting. The break is editorial.

Qualifications noted at the head of the 4. Where third-party code sits inside the product apply to these rows equally, and are not restated for each part. The parts are to be read together.

Schedule O - Robot variant, module revision and bench rig coverage (TDR-2026-07) (continued, part 2 of 2)

Robot variant	Module	Firmware revision	Sites deployed	Bench rig coverage	Revision status
HR-240 tote mover	navigation lidar	4.5.2	11	rig one only	current
HR-300 pallet mover	load cell deck	1.16.7	3	rig two only	superseded
HR-240 tote mover	E-stop harness	1.5.3	13	both rigs	one revision behind
HR-300 pallet mover	navigation lidar	2.8.3	7	both rigs	current

Robot variant	Module	Firmware revision	Sites deployed	Bench rig coverage	Revision status
HR-320 pallet mover	load cell deck	1.8.6	4	neither rig	superseded
HR-300 pallet mover	navigation lidar	4.17.7	6	both rigs	current
HR-410 manipulator	load cell deck	3.11.2	3	rig two only	one revision behind
HR-410 manipulator	safety scanner	4.2.4	13	rig two only	pre-production
HR-320 pallet mover	safety scanner	3.8.3	12	neither rig	superseded
HR-410 manipulator	safety scanner	3.13.5	12	both rigs	current
HR-320 pallet mover	E-stop harness	3.13.2	5	rig one only	superseded
HR-240 tote mover	safety scanner	3.9.3	2	rig one only	one revision behind
HR-320 pallet mover	safety scanner	1.11.7	2	neither rig	pre-production
HR-240 tote mover	network radio	3.1.3	13	neither rig	current
HR-200 tote mover	gripper coprocessor	3.17.0	10	rig one only	pre-production
HR-410 manipulator	charge contact assembly	3.8.1	4	rig two only	superseded
HR-320 pallet mover	load cell deck	3.18.3	4	rig one only	current
HR-200 tote mover	safety scanner	4.11.6	6	both rigs	superseded
HR-320 pallet mover	charge contact assembly	1.19.5	10	both rigs	pre-production
HR-320 pallet mover	inertial measurement unit	2.2.5	5	rig one only	one revision behind
HR-240 tote mover	E-stop harness	1.12.1	8	both rigs	superseded
HR-300 pallet mover	charge contact assembly	2.7.7	2	rig two only	pre-production

5. Provenance, approval and contributor assignment

Pgbouncer-managed carries a practical obligation: to share alike any derived asset it distributes. The supply route for pgbouncer-managed is public registry, signature verified. Pgbouncer-managed carries ISC terms, classified no for copyleft purposes. pgbouncer-managed is the entry that changed most between the first and second document requests behind TDR-2026-07.

Media-transcode obliges the company to share alike any derived asset it distributes. Media-transcode presents a component with open advisories that do not affect the deployed configuration to the engineering team. Approval for media-transcode reads approved with conditions 2024-08-16, and the reviewer checked the approving record. media-transcode illustrates why TDR-2026-07 separates the severity of a finding from the confidence behind it.

Grid-render was assessed for support as well as licence risk, and it carries a single-maintainer dependency with no succession plan. Grid-render carries the approval entry approved with conditions 2023-07-19; the reviewer looked for a corresponding decision record. The operational obligation attaching to grid-render is to share alike any derived asset it distributes. grid-render sits inside the sample rather than being an exception to it, which is the reason it is reported at length.

On provenance, lidar-vendor-sdk shows supplied by the sensor vendor. On lidar-vendor-sdk the reviewer records that two major versions behind current; the vendor's support statement covers the deployed version to 2027. Lidar-vendor-sdk version 4.1.2 is catalogued at OSS-018 and is used lidar driver. lidar-vendor-sdk is the only entry in its group where the reviewer and the engineering team reached different conclusions.

Yaml-parse also carries a build-time dependency whose absence would stop the pipeline rather than the product, which is a supportability question rather than a licence one. Attribution present is the reviewer's observation on yaml-parse. Policy compliance for yaml-parse rests on its approval entry, which is approved 2023-09-29. yaml-parse was tested rather than accepted, and the test is described in the working papers behind TDR-2026-07.

The licence on k8s-operator-framework is Apache-2.0; copyleft exposure is no. K8s-operator-framework sits inside Site Provisioning Service and is therefore in the shipped product rather than in tooling. Public registry, signature verified is the provenance recorded for k8s-operator-framework. k8s-operator-framework is the only entry in its group where the reviewer and the engineering team reached different conclusions.

Linking mode matters for otel-instrument, which is used tracing across cloud services. The licence on otel-instrument is Apache-2.0; copyleft exposure is no. Otel-instrument appears at OSS-023 in version 1.24.0; the usage recorded is tracing across cloud services. otel-instrument was reviewed against the position twelve months earlier, and the direction of travel is favourable.

Wms-adapter-sdk version 2.2.0 is catalogued at OSS-017 and is used one customer's warehouse system adapter. To exclude a particular build configuration from the shipped artefact is what wms-adapter-sdk requires in practice. Because wms-adapter-sdk is used one customer's warehouse system adapter, its licence terms reach the delivered software. wms-adapter-sdk was the subject of a specific management representation, which the reviewer has recorded but not relied on alone.

OSS-002 records geom2d-fusion 0.9.7 against Costmap Fusion, dynamically linked by the fusion pipeline. Geom2d-fusion is recorded as approved 2023-08-04 in the component approval log. Where geom2d-fusion came from matters here: public registry; upstream branch unmaintained since 2024. geom2d-fusion is recorded here without adjustment; the reviewer saw no basis to restate what the company reported.

To attribute the component in the customer-facing notice file is what rtos-shim requires in practice. Rtos-shim is licensed under BSD-2-Clause, with copyleft exposure recorded as no. Rtos-shim presents an upstream project that is active but changes interface frequently to the engineering team. rtos-shim is the clearest instance in TDR-2026-07 of a control that exists on paper and is weaker in practice.

Schedule U - Qualifications carried forward, with evidence strength (TDR-2026-07)

Ref	Qualification carried forward	Evidence strength	Effect on conclusion	What would close it
OQ-15-01	the compensating control was not tested	limited	narrows the conclusion to the tested scale	one further interview
OQ-15-02	the population completeness could not be proven	limited	would change the conclusion if resolved adversely	a document the company has not produced
OQ-15-03	the remediation landed without a regression test	moderate	narrows the conclusion to the tested scale	a witnessed test
OQ-15-04	the remediation landed without a regression test	limited	is reflected in the post-close plan	a document the company has not produced
OQ-15-05	the supplier certificate was read rather than the test witnessed	limited	would change the conclusion if resolved adversely	a witnessed test
OQ-15-06	the measurement basis changed during the reviewed period	moderate	is reflected in the post-close plan	a decision from management
OQ-15-07	the retention window closed before the raw data was requested	moderate	is reflected in the post-close plan	a decision from management
OQ-15-08	the objective is met on the company's own measurement only	strong	is reflected in the post-close plan	a re-run of the original test
OQ-15-09	the measurement basis changed during the reviewed period	strong	does not change the conclusion	one further interview
OQ-15-10	the objective is met on the company's own measurement only	limited	does not change the conclusion	a decision from management
OQ-15-11	the objective is met on the company's own measurement only	limited	is reflected in the post-close plan	a document the company has not produced
OQ-15-12	the retention window closed before the raw data was requested	strong	would change the conclusion if resolved adversely	a decision from management
OQ-15-13	the measurement basis changed during the reviewed period	strong	would change the conclusion if resolved adversely	an independent measurement
OQ-15-14	the measurement basis changed during the reviewed period	moderate	does not change the conclusion	one further interview
OQ-15-15	the measurement basis changed during the reviewed period	limited	narrows the conclusion to the tested scale	a document the company has not produced
OQ-15-16	the measurement basis changed during the reviewed period	moderate	is reflected in the post-close plan	a re-run of the original test
OQ-15-17	the supplier certificate was read rather than the test witnessed	moderate	would change the conclusion if resolved adversely	a witnessed test

Ref	Qualification carried forward	Evidence strength	Effect on conclusion	What would close it
OQ-15-18	the remediation landed without a regression test	strong	is reflected in the post-close plan	one further interview
OQ-15-19	the declined item has no expiry date	strong	leaves one section unconcluded	a re-run of the original test
OQ-15-20	the second bench rig does not cover this path	strong	does not change the conclusion	one further interview
OQ-15-21	the objective is met on the company's own measurement only	limited	is reflected in the post-close plan	a re-run of the original test
OQ-15-22	the compensating control was not tested	moderate	is reflected in the post-close plan	a re-run of the original test
OQ-15-23	the declined item has no expiry date	moderate	would change the conclusion if resolved adversely	one further interview
OQ-15-24	the population completeness could not be proven	strong	does not change the conclusion	a decision from management
OQ-15-25	the measurement basis changed during the reviewed period	moderate	narrows the conclusion to the tested scale	an independent measurement
OQ-15-26	the compensating control was not tested	moderate	would change the conclusion if resolved adversely	a decision from management
OQ-15-27	the remediation landed without a regression test	strong	is reflected in the post-close plan	a witnessed test
OQ-15-28	the second bench rig does not cover this path	moderate	does not change the conclusion	one further interview
OQ-15-29	the population completeness could not be proven	moderate	does not change the conclusion	one further interview
OQ-15-30	the second bench rig does not cover this path	moderate	leaves one section unconcluded	a witnessed test
OQ-15-31	the supplier certificate was read rather than the test witnessed	moderate	narrows the conclusion to the tested scale	a document the company has not produced
OQ-15-32	the declined item has no expiry date	limited	leaves one section unconcluded	a witnessed test

6. Findings arising from dependency and secret scanning

Finding SEC-2025-02 arrives as a High finding with a 8.1 score, and the reviewer did not adjust either. Evidence behind SEC-2025-02 is that the tester extracted a working token from a retained log, which the reviewer treats as moderate, because the demonstration was performed in a laboratory tenant rather than in production. Remediated is the status of Release Orchestrator finding SEC-2025-02, dated 2025-09-19. 2025-09-17 is the discovery date of High-severity finding SEC-2025-02, which finds that build logs printed the full authorization header. Release Orchestrator finding SEC-2025-02 is recorded here without adjustment; the reviewer saw no basis to restate what the company reported.

On SEC-2026-17 against Motion Planner the tester's evidence was that override applied on a laboratory robot and observed in plan output, and the reviewer grades it strong. Weighing High-severity finding SEC-2026-17 against the compensating controls, the remediation is adequate and the time to remediate is not. 2026-06-19 is the closure position on Motion Planner finding SEC-2026-17, which the ledger records as remediated. High-severity finding SEC-2026-17 sits against Motion Planner and records that motion planner parameter override accepted from an unauthenticated local port. SEC-2026-17 is the only entry in its group where the reviewer and the engineering team reached different conclusions.

Residual risk after remediation of SEC-2026-18 is low; no fleet impact demonstrated. The tester scored assessment finding SEC-2026-18 at 3.4 and labelled it Low. Reading finding SEC-2026-18, the defect is that battery pack diagnostic interface has no rate limit; the discovery date is 2026-04-23. Assessment finding SEC-2026-18 is assessed as follows: the item is open past its committed date and no revised date has been agreed. SEC-2026-18 was checked against the corresponding entry in the company's own register, and the register was current.

Tracking finding SEC-2026-16 to closure gives a status of remediated and a date of 2026-06-26. The reviewer's position on Medium-severity finding SEC-2026-16 is that the exposure is internal rather than customer-facing, which lowers the severity without removing the finding. Rating for finding SEC-2026-16 is Medium, band 5.8; the reviewer re-

read the scoring rationale and accepted it. What remains after Medium-severity finding SEC-2026-16 is remediated is low; the socket binds to the loopback interface only. SEC-2026-16 against Costmap Fusion is one of the items the reviewer would put on a monitoring pack rather than on a closing checklist.

Traffic Manager finding SEC-2026-14 stands remediated; 2026-06-18 is what the record gives for closure. The practical consequence in SEC-2026-14 against Traffic Manager is that a manual corridor override cannot be attributed after the fact. SEC-2026-14 reduces to a residual risk of low. The reviewer declines to soften Traffic Manager finding SEC-2026-14: the compensating control is real but depends on a network boundary that a single misconfiguration removes. finding SEC-2026-14 was discussed with two engineers who gave slightly different accounts, and the difference is preserved here.

Strong evidence supports assessment finding SEC-2026-15, namely that collision created in a staging tenant. Remediation status on Site Provisioning Service finding SEC-2026-15 is accepted; the record shows not planned. Rating for SEC-2026-15 against Site Provisioning Service is Low, band 3.0; the reviewer re-read the scoring rationale and accepted it. SEC-2026-15 is assessed as follows: the item is open past its committed date and no revised date has been agreed. Low-severity finding SEC-2026-15 is the entry that changed most between the first and second document requests behind TDR-2026-07.

SEC-2025-05 against Telemetry Ingest produces this judgement: the compensating control is real but depends on a network boundary that a single misconfiguration removes. High-severity finding SEC-2025-05 reduces to a residual risk of low; attestation is now required on the ingest path. The finding at SEC-2025-05, dated 2025-09-19, is that telemetry ingest accepted a robot identity without checking device attestation. What High-severity finding SEC-2025-05 enables is this: a forged agent could inject telemetry attributed to a real robot. Telemetry Ingest finding SEC-2025-05 is recorded here without adjustment; the reviewer saw no basis to restate what the company reported.

On SEC-2026-01 against Release Orchestrator the conclusion is that the underlying pattern appears in two other components that the tester did not reach. Limited, resting on a configuration read rather than on a demonstrated exploitation evidence supports Release Orchestrator finding SEC-2026-01, namely that the tester used the path in a staging tenant and the audit trail recorded no justification. The residual position on SEC-2026-01 against Release Orchestrator is high until the second-approver control lands, stated after mitigation rather than assuming full success. The tester scored High-severity finding SEC-2026-01 at 7.1 and labelled it High. Release Orchestrator finding SEC-2026-01 was inspected in the repository rather than described from the ticket, which is why the account is specific.

As at the review date Localization Engine finding SEC-2026-19 was open and the closure entry reads committed 2026-10-15. On 2026-04-23 the tester recorded Localization Engine finding SEC-2026-19: localization map cache stored unencrypted on the robot. For assessment finding SEC-2026-19 the record shows that cache read from a removed module, which is moderate evidence and is graded separately from severity. SEC-2026-19 was graded Medium, band 5.1, by Meridian Assurance engagement MA-2026-1. Localization Engine finding SEC-2026-19 carries a follow-up request that was satisfied after the draft of TDR-2026-07 was circulated.

Finding SEC-2026-03 rests on the observation that the tester applied an unsigned bundle on a laboratory robot; the reviewer holds that to be moderate. Remediation status on High-severity finding SEC-2026-03 is remediated; the record shows 2026-06-12. 2026-04-15 is the discovery date of finding SEC-2026-03, which finds that config bundle signature verification could be disabled by a local flag. High-severity finding SEC-2026-03 reduces to a residual risk of low; the flag was removed and the agent refuses unsigned bundles. assessment finding SEC-2026-03 was raised again with the owning engineer after the initial interview, and the position did not change.

The residual position on Medium-severity finding SEC-2025-06 is low; the index is scoped per tenant and the identifiers are opaque, stated after mitigation rather than assuming full success. Medium-severity finding SEC-2025-06 is remediated, with a closure date of 2025-12-18. Medium-severity finding SEC-2025-06 translates into the following exposure: a customer operator could enumerate the existence of clips in another tenant. Medium-severity finding SEC-2025-06 rests on the observation that enumeration confirmed; no clip content was retrievable; the reviewer holds that to be moderate, because the demonstration was performed in a laboratory tenant rather than in production. SEC-2025-06 against Video Snippet Service sits inside the sample rather than being an exception to it, which is the reason it is reported at length.

Tracking Alarm Router finding SEC-2026-05 to closure gives a status of open and a date of committed 2026-09-30. Assessment finding SEC-2026-05 arrives as a Medium finding with a 5.2 score, and the reviewer did not adjust either. SEC-2026-05 reduces to a residual risk of medium; per-region credentials are scheduled but not issued. What SEC-2026-05 against Alarm Router enables is this: one leaked credential affects paging for every region. SEC-2026-05 is reported here in full because a summary of it in an earlier draft of TDR-2026-07 was misread as an assurance.

2025-09-25 is the discovery date of SEC-2025-11 against Remote Operations Console, which finds that console session cookie lacked a strict same-site attribute. On Low-severity finding SEC-2025-11 the tester's evidence was that attribute absence confirmed; no working exploit chain was demonstrated, and the reviewer grades it limited, resting on a configuration read rather than on a demonstrated exploitation. Weighing assessment finding SEC-2025-11 against the compensating controls, the finding was closed within the committed window and the closure evidence was inspected. Remediation status on SEC-2025-11 against Remote Operations Console is remediated; the record shows 2025-10-30. Low-severity finding SEC-2025-11 rests on a single source, and the reviewer has said so rather than implying corroboration.

What Medium-severity finding SEC-2026-04 enables is this: customer commercial data is retained in operational logs. SEC-2026-04 records that wMS connector logged customer order payloads at debug level, discovered on 2026-04-16. Medium-severity finding SEC-2026-04 is not fully retired by its remediation: low; payload logging requires an explicit per-tenant approval. On SEC-2026-04 against WMS Connector the conclusion is that management's acceptance is defensible on the evidence and should be revisited before the fleet doubles. SEC-2026-04 against WMS Connector is recorded here without adjustment; the reviewer saw no basis to restate what the company reported.

7. Remediation of the licence exposure and its cost

For quality item TD-04 the team estimates 2 engineer-quarters, which the reviewer converted to $240,000 for comparison. In the programme Costmap Fusion item TD-04 is behind a customer maintenance window that is not yet agreed. TD-04 is the item to raise robot software line coverage above sixty per cent on the perception path, classified under quality. quality item TD-04 is reported at the evidence strength available and has not been promoted to fit the narrative.

Sequencing of TD-09 matters: it is in parallel with the release-process changes. The reviewer's comment on programme item TD-09 is that the upgrade requires a maintenance window at every site. Funding for TD-09 on Event Bus is funded and already staffed, and the register does not say so. TD-09 on Event Bus is the clearest instance in TDR-2026-07 of engineering practice being better than the documentation suggests.

Performance item TD-13 is sequenced behind a customer maintenance window that is not yet agreed. Effort on programme item TD-13 is estimated at 1.5 engineer-quarters, costed at $178,000. On Localization Engine item TD-13 the reviewer records that carries the same dependency as the copyleft removal. TD-13 on Localization Engine rests on a single source, and the reviewer has said so rather than implying corroboration.

Reading the programme plan, TD-05 is behind a platform change that has its own target date. Under test capability, entry Hardware-in-the-Loop Rig item TD-05 is to bring the second bench rig to the current sensor configuration. Exposure carried by programme item TD-05 is that it one sensor path is verified on a single rig. test capability item TD-05 is the clearest instance in TDR-2026-07 of a control that exists on paper and is weaker in practice.

$92,000 is the costed estimate for TD-23 on Charge Scheduler, based on 0.75 engineer-quarters. The business consequence recorded against programme item TD-23 is that it one noisy query affects task assignment. In the programme register item TD-23 is behind a supplier release outside the company's control. Charge Scheduler item TD-23 was cross-checked against the engineering record supplied for TDR-2026-07, and the two agree.

Register item TD-10 is ahead of two other items that depend on it, which is why the target date is later than the effort estimate implies. If register item TD-10 is not done, the exposure is that it the analytics load fails when the function is withdrawn. The estimate for register item TD-10 is 0.5 engineer-quarters and $54,000. register item TD-10 was cross-checked against the engineering record supplied for TDR-2026-07, and the two agree.

About register item TD-21 it should be said that data collection is running at the two affected sites. TD-21 cannot start earlier than it does because it is behind a platform change that has its own target date. TD-21 on Battery

Management Service is partly funded, with the second phase unbudgeted in the engineering plan the reviewer was shown. Battery Management Service item TD-21 is the entry that changed most between the first and second document requests behind TDR-2026-07.

As at the review date safety item TD-19 was in progress, with 2026-08-31 still the stated target. $268,000 is the costed estimate for safety item TD-19, based on 2 engineer-quarters. Speed and Separation Monitor item TD-19 is unfunded and dependent on capital rather than engineering effort; the register status and the funding position are not the same thing. programme item TD-19 was checked against the company's own register, which had not been updated since the change landed.

The business consequence recorded against TD-20 on Wheel Odometry Filter is that it drift on wet floors is neither corrected nor flagged. Register item TD-20 covers the work to validate and ship the wheel slip detector. The reviewer's qualification on TD-20 on Wheel Odometry Filter is that the prototype uses a network-copyleft component that must be replaced before shipping. TD-20 on Wheel Odometry Filter was discussed with two engineers who gave slightly different accounts, and the difference is preserved here.

On quality item TD-06 the reviewer records that blocked on the physics core determinism work. The consequence of leaving programme item TD-06 undone is that it a nightly regression can reach a release candidate. The register shows Simulation Harness item TD-06 as blocked, due 2026-11-30. TD-06 sits inside the sample rather than being an exception to it, which is the reason it is reported at length.

The estimate for programme item TD-18 is 0.75 engineer-quarters and $104,000. Funding for safety evidence item TD-18 is funded only if the team's headcount plan is met, and the register does not say so. Exposure carried by Safety Controller item TD-18 is that it investigation evidence is lost between weekly extractions. safety evidence item TD-18 is not material on its own and belongs in this report because of what it reveals about the surrounding process.

The dependency structure places security supportability item TD-08 last in a three-item chain, so its date is the least reliable on the register. TD-08 covers the work to replace the archived charting library in the operator console. Register item TD-08 is funded from the second quarter onward in the engineering plan the reviewer was shown. register item TD-08 would move from a concern to a verified defect if one further measurement were supplied.

Sizing on register item TD-11 gives 0.5 engineer-quarters and $61,000. Configuration item TD-11 is on the register because it template drift produced a go-live failure in May 2026. Site Provisioning Service item TD-11 appears in the register and is not funded, and carried in the register regardless. TD-11 was reviewed against the position twelve months earlier, and the direction of travel is favourable.

TD-22 is on the register because it calibration is lost on a compute module swap. As at the review date register item TD-22 was backlog, with 2027-01-31 still the stated target. Quality item TD-22 is ahead of two other items that depend on it, which is why the target date is later than the effort estimate implies. programme item TD-22 was checked against the company's own register, which had not been updated since the change landed.

8. Incidents involving a third-party component

Observability is the class of Edge Gateway Agent incident INC-2026-045; Edge Gateway Agent is the component. The engineering team's root cause for Evergreen Warehousing incident INC-2026-045 is that the buffer is sized for four hours and drops oldest-first without raising an alert, and the reviewer found the analysis sound. Contributing to observability incident INC-2026-045, and not repaired by the remediation, is that buffer pressure was never aggregated to a site view, so the condition was not observable. Attention came to observability event INC-2026-045 when the buffer pressure metric was visible per robot but no site-level alert existed. Observability event INC-2026-045 was contained to one site and 39 robots. Evergreen Warehousing incident INC-2026-045 is reported at the evidence strength available and has not been promoted to fit the narrative.

Incident INC-2026-045 carries limited; the log retention window had closed before the reviewer asked for the raw data evidence, and the reviewer declined to raise the finding beyond what that supports. Evergreen Warehousing incident INC-2026-045 was raised on 2026-04-21 after a site-wide network change left robots buffering for five hours and the oldest telemetry samples were discarded. Incident INC-2026-045 took 5 h 12 m to stabilise and 26 days to close out. Evergreen Warehousing incident INC-2026-045 produced one commitment: to aggregate buffer pressure to a site alert

and extend the buffer to twelve hours on the current hardware, owned by Sam Oyelaran, dated 2026-06-30. The alert landed on 2026-05-17; the buffer extension is in progress is the position on the incident INC-2026-045 commitment as at the review date. observability event INC-2026-045 illustrates why TDR-2026-07 separates the severity of a finding from the confidence behind it.

Strong on the cause and limited on the contributing factor is the evidence grade for incident INC-2026-021, and the reader should not read severity into it. The reviewer treats one contributing factor in tooling incident INC-2026-021 as the more important lesson: the build pinned the compiler but not the optimisation flags used for release builds. Colin Achebe committed after incident INC-2026-021 to pin the full build configuration and add a determinism check to the harness, due 2026-05-31. Simulation Harness owns incident INC-2026-021, which the reviewer classes as tooling. Mitigation of tooling event INC-2026-021 took not applicable; developer tooling and full resolution took open at the review date. tooling event INC-2026-021 carries a follow-up request that was satisfied after the draft of TDR-2026-07 was circulated.

Cause analysis in INC-2026-021 concluded that the physics core is not deterministic across optimisation levels. Simulation Harness incident INC-2026-021 is recorded as follows on delivery: not landed; the determinism check is written but not enabled. On 2026-02-24 the record for internal incident INC-2026-021 opens: a nightly simulation suite produced non-reproducible results after the physics core was rebuilt. So far as reach is concerned, internal incident INC-2026-021 involved no production site and no robot directly, the exposure being to internal systems. Simulation Harness incident INC-2026-021 is not read as an isolated failure: the team's analysis was honest about a gap it could have concealed. Simulation Harness incident INC-2026-021 was discussed with two engineers who gave slightly different accounts, and the difference is preserved here.

The judgement recorded against incident INC-2026-009 is that the commitment date was met and the verifying test now exists, which is not the pattern across the whole register. Beyond the direct cause, INC-2026-009 shows that the service workflow allowed the step to be marked complete without a calibration record. On delivery, Gripper Controller incident INC-2026-009 shows this: landed on 2026-03-16 with a workflow gate. For calibration incident INC-2026-009 the reviewer holds moderate on the cause and strong on the delivery position, which was checked in the repository evidence and reports the finding at that strength. After INC-2026-009, Ines Duarte took the action to require a calibration record before returning a manipulator robot to service by 2026-04-15. calibration event INC-2026-009 is not material on its own and belongs in this report because of what it reveals about the surrounding process.

Time to mitigate incident INC-2026-009 was 1 h 30 m to quarantine the affected robots, and time to resolve was 48 days. 2026-01-27 is the date of calibration incident INC-2026-009, on which a gripper applied more force than the configured limit after a mechanical replacement without recalibration. Gripper Controller accounts for Gripper Controller incident INC-2026-009, a calibration event. How Blue Mesa Components incident INC-2026-009 surfaced matters to the assessment: the payload damage was reported by the customer. What went wrong in calibration incident INC-2026-009 was that force limiting depends on a calibration that is only triggered by a service workflow step. INC-2026-009 is reported here in full because a summary of it in an earlier draft of TDR-2026-07 was misread as an assurance.

Timing event INC-2025-047 touched one site and 12 robots, and Asterion Logistics was the affected account. Yuki Harada committed after timing incident INC-2025-047 to re-derive the time offset on every reconnect and add the affected variant to the second bench rig, due 2025-06-15. Timing event INC-2025-047 supports one conclusion and not another: the blast radius was set by a deployment topology decision made years earlier. Lidar Driver incident INC-2025-047 was caused by a design gap: the driver did not re-derive the sensor time offset on reconnect. The driver fix landed on 2025-04-05; the second rig was still unmodified at review is the position on the Asterion Logistics incident INC-2025-047 commitment as at the review date. incident INC-2025-047 is the entry that changed most between the first and second document requests behind TDR-2026-07.

Timing incident INC-2025-047 carries a second-order finding: the bench rig that would have caught it carries a different lidar variant. Attention came to Lidar Driver incident INC-2025-047 when an automated fusion timing alert fired inside four minutes. Incident INC-2025-047 shows a short mitigation and a long tail: 38 m against 3 days. Grading timing incident INC-2025-047 on evidence rather than on consequence gives moderate; the narrative is consistent but the reviewer could not independently reproduce it. The classification applied to Asterion Logistics incident INC-2025-047 is timing, and the affected component is Lidar Driver. timing incident INC-2025-047 is the reason one section of TDR-2026-07 concludes narrowly rather than generally.

Process control event INC-2026-062 supports one conclusion and not another: the remediation addresses the immediate cause and leaves the contributing factor in place. Following INC-2026-062 the team undertook to require a justification and a second approver within one working day for every emergency release, with Colin Achebe accountable by 2026-08-31. The engineering team's root cause for process control event INC-2026-062 is that the emergency path is available to any release engineer without a justification field, and the reviewer found the analysis sound. 2026-06-02 is the date of incident INC-2026-062, on which an emergency release path with a single approver was used for a change that was not an emergency. The affected population in incident INC-2026-062 was no production site and no robot directly, the exposure being to internal systems. INC-2026-062 is the entry that changed most between the first and second document requests behind TDR-2026-07.

Process control incident INC-2026-062 carries a second-order finding: the policy defines an emergency but the tool does not ask which condition applies. Not landed; the change is scoped but not scheduled is the position on the process control incident INC-2026-062 commitment as at the review date. Mitigation of Release Orchestrator incident INC-2026-062 took not applicable; process only and full resolution took open at the review date. Detection came through the release audit found it during the quarterly review, which places INC-2026-062 on the reported side of the ledger. Process control is the class of process control event INC-2026-062; Release Orchestrator is the component. process control incident INC-2026-062 was raised again with the owning engineer after the initial interview, and the position did not change.

Contributing to INC-2025-084, and not repaired by the remediation, is that the pause control had been added in a hotfix without a design review. Assessed on its own record, Blue Mesa Components incident INC-2025-084 shows that management's severity rating is defensible and its remediation priority is not. For state persistence event INC-2025-084 the reviewer holds limited, being a single account corroborated only by the ticket timeline evidence and reports the finding at that strength. Incident INC-2025-084 records that on 2025-07-08, a paused update ring resumed after a routine service restart and eleven robots took an unintended firmware step. In OTA Updater incident INC-2025-084 the first signal was the site team escalated when robots rebooted outside the maintenance window, and the reviewer treats that as reported detection. state persistence event INC-2025-084 was discussed with two engineers who gave slightly different accounts, and the difference is preserved here.

Cause analysis in state persistence incident INC-2025-084 concluded that the pause flag was held in process memory rather than in the release state record. OTA Updater owns incident INC-2025-084, which the reviewer classes as state persistence. Time to mitigate OTA Updater incident INC-2025-084 was 27 m, and time to resolve was 5 days. Following INC-2025-084 the team undertook to persist the pause state and require a design note for any change to the release state machine, with Erik Palmgren accountable by 2025-08-15. State persistence event INC-2025-084 was contained to one site and 11 robots. OTA Updater incident INC-2025-084 is one of the entries where TDR-2026-07 relies on the company's own measurement rather than an independent one.

Verification of the secret exposure incident INC-2025-118 remediation gives redaction landed on 2025-09-19; retention was changed on 2025-09-24. In incident INC-2025-118 the first signal was the dependency and secret scanner flagged it on the following scan, and the reviewer treats that as alerted detection. The engineering team's root cause for secret exposure event INC-2025-118 is that a debug logging statement in the connector build printed the full authorization header, and the reviewer found the analysis sound. A contributing factor in secret exposure incident INC-2025-118 was that build logs were retained far longer than the token lifetime, widening the exposure window. The two durations recorded for internal incident INC-2025-118 are 3 h 40 m to revoke and rotate to mitigate and 7 days to resolve. secret exposure incident INC-2025-118 would be a candidate for the first post-close engineering review under any reasonable plan.

In taxonomy terms incident INC-2025-118 is a secret exposure incident in Identity and Authorization. Incident INC-2025-118 affected no production site and no robot directly, the exposure being to internal systems. Secret exposure incident INC-2025-118 produced one commitment: to redact authorization headers at the logging layer and cut build log retention to fourteen days, owned by Sonia Krall, dated 2025-10-31. INC-2025-118 records that on 2025-09-17, a service token with a twelve-hour lifetime was found in a build log retained for ninety days. Assessed on its own record, incident INC-2025-118 shows that an operational workaround masked the defect for longer than the incident record suggests. incident INC-2025-118 illustrates why TDR-2026-07 separates the severity of a finding from the confidence behind it.

INC-2025-103 affected one site and 2 robots. Motion Planner incident INC-2025-103 produced one commitment: to add a deterministic tie-break and return the pinch-point scenario to the nightly suite, owned by Priya Raghunathan, dated 2025-10-31. Beyond the direct cause, algorithm defect event INC-2025-103 shows that the simulation scenario for this geometry existed but had been excluded from the nightly suite. The tie-break landed on 2025-10-19; the scenario was reinstated on 2025-09-02 is the position on the Motion Planner incident INC-2025-103 commitment as at the review date. The engineering team's root cause for Ironwood Grocery Group incident INC-2025-103 is that the yield heuristic had no tie-break and both robots computed the same deference score, and the reviewer found the analysis sound. incident INC-2025-103 carries a follow-up request that was satisfied after the draft of TDR-2026-07 was circulated.

Mitigation of Ironwood Grocery Group incident INC-2025-103 took 19 m and full resolution took 58 days. Incident INC-2025-103 supports one conclusion and not another: management's severity rating is defensible and its remediation priority is not. Evidence strength on the Ironwood Grocery Group incident INC-2025-103 analysis is moderate on the cause and strong on the delivery position, which was checked in the repository. Incident INC-2025-103 reached the engineering team when the queue depth alert introduced after the March incident fired correctly. In taxonomy terms Ironwood Grocery Group incident INC-2025-103 is a algorithm defect incident in Motion Planner. incident INC-2025-103 has an owner who accepted the finding immediately, which is worth recording alongside the finding itself.

9. Performance dependence on third-party components

LT-2026-01 supports this conclusion: the test environment carried a smaller dataset than production, so the database term is understated here. The applicable target for scenario LT-2026-01 is the thirty-second telemetry visibility objective, and the reviewer checked where that target came from. The test bed for 720-robot run LT-2026-01 carries a synthetic robot fleet driven from recorded mission traces, which is the main reason the reviewer does not treat it as production-equivalent. scenario LT-2026-01 was cross-checked against the engineering record supplied for TDR-2026-07, and the two agree.

Measured latency in 960-robot run LT-2026-02 was 455 ms at the median, 1,340 ms at p95 and 2,880 ms at p99. The applicable target for LT-2026-02 is the five-minute configuration distribution objective, and the reviewer checked where that target came from. Scenario LT-2026-02 used the standard performance environment, which is characterised by a replayed order stream taken from the two largest sites. run LT-2026-02 was inspected in the repository rather than described from the ticket, which is why the account is specific.

The size of the concurrent operator session pool is what separates eight sites with the allocator read replicas enabled run LT-2026-03 from the adjacent run. In 960-robot run LT-2026-03 the median was 400 ms and the tail reached 2,310 ms; errors ran at 0.21%. The result recorded for run LT-2026-03 is pass, which the reviewer traced to the underlying measurement. eight sites with the allocator read replicas enabled run LT-2026-03 was cross-checked against the engineering record supplied for TDR-2026-07, and the two agree.

Scenario LT-2026-04 ran against an environment with the current production topology with one region held out. Load run LT-2026-04 exists in the series to isolate the warehouse system endpoint used. The result recorded for load run LT-2026-04 is fail against the 1,200 ms p95 target, which the reviewer traced to the underlying measurement. 1200-robot run LT-2026-04 illustrates why TDR-2026-07 separates the severity of a finding from the confidence behind it.

Measurement of load run LT-2026-05 is against the fifteen-minute map publication objective rather than against a customer commitment. Comparing scenario LT-2026-05 with its neighbour in the series, the difference is the warehouse system endpoint used. Configuration of the environment for load run LT-2026-05 included production data volumes and a stubbed paging provider. order burst after the broker queue change run LT-2026-05 was reviewed against the position twelve months earlier, and the direction of travel is favourable.

The shard rebalance under peak load run at shard rebalance under peak load run LT-2026-06 was executed on 2026-02-19 at 720 robots and 13,800 orders an hour. The target 720-robot run LT-2026-06 was measured against is the 0.5 per cent error-rate ceiling in the internal service documentation. What LT-2026-06 does and does not establish is worth stating: the reviewer would repeat this run with the newest site's order profile before relying on it. 720-robot run LT-2026-06 is reported at the evidence strength available and has not been promoted to fit the narrative.

Identity service token storm run LT-2026-07 supports this conclusion: the measurement is credible and the extrapolation the company draws from it is not. The limiting term in load run LT-2026-07 was token issuance was not a

limiting term at any tested scale. Scenario LT-2026-07 is the identity service token storm run, sized at 960 robots and 18,200 orders an hour. 960-robot run LT-2026-07 is one of the entries where TDR-2026-07 relies on the company's own measurement rather than an independent one.

Run LT-2026-08 exists in the series to isolate whether control and telemetry shared a topic. In console session concurrency run LT-2026-08 the bottleneck was tested to 140 concurrent operator sessions. 480-robot run LT-2026-08 is not read as a headline number: the test environment carried a smaller dataset than production, so the database term is understated here. LT-2026-08 was the subject of a specific management representation, which the reviewer has recorded but not relied on alone.

480-robot run LT-2026-09 is the console session concurrency at twice the plan run, sized at 480 robots and 9,400 orders an hour. Scenario LT-2026-09 produced a 380 ms median with 1,460 ms and 3,900 ms tails and an 0.52% error rate. Against the stated targets, LT-2026-09 is fail. scenario LT-2026-09 is the entry that changed most between the first and second document requests behind TDR-2026-07.

On 2026-02-26 the team ran scenario LT-2026-10, the warehouse connector burst, one tenant case, at 240 robots. 1,830 ms at p95 is the figure that matters in 240-robot run LT-2026-10, against 610 ms median and 4,400 ms at the tail. The size of the concurrent operator session pool is what separates run LT-2026-10 from the adjacent run. scenario LT-2026-10 is one of the items the reviewer would put on a monitoring pack rather than on a closing checklist.

The applicable target for scenario LT-2026-11 is the 1,200 ms p95 objective published for order assignment, and the reviewer checked where that target came from. Set beside the other runs, scenario LT-2026-11 varies only in whether the read replicas were enabled. Pass is what the test record gives for run LT-2026-11. load run LT-2026-11 is the reason one section of TDR-2026-07 concludes narrowly rather than generally.

The ingest path scales horizontally with no shared bottleneck was the constraint LT-2026-12 exposed. The judgement the reviewer draws from LT-2026-12 is that the failure is a capacity finding rather than a defect, and the remediation is a scheduled engineering programme. The test bed for run LT-2026-12 carries a fleet simulator rather than physical robots, with the control loop unchanged, which is the main reason the reviewer does not treat it as production-equivalent. run LT-2026-12 is the clearest instance in TDR-2026-07 of engineering practice being better than the documentation suggests.

The judgement the reviewer draws from load run LT-2026-13 is that the result holds only for the tested order mix, which is drawn from two sites and is not representative of the newest ones. Scenario LT-2026-13 is the twelve sites, projected 2027 fleet run, sized at 1440 robots and 27,000 orders an hour. The fifteen-minute map publication objective is the target applied to LT-2026-13. scenario LT-2026-13 has an owner who accepted the finding immediately, which is worth recording alongside the finding itself.

Scenario LT-2026-14 exercised the four hour edge buffer drain after an outage case with 720 robots and 13,800 orders an hour, on 2026-04-15. Set beside the other runs, scenario LT-2026-14 varies only in the size of the concurrent operator session pool. The environment behind LT-2026-14 carries a replayed order stream taken from the two largest sites, and the reviewer weighted the result accordingly. four hour edge buffer drain after an outage run LT-2026-14 was excluded from an earlier draft of TDR-2026-07 and reinstated after the reviewer obtained the supporting record.

10. Licence questions left open at the review date

Where the report at TDR-2026-07 cannot conclude, it says what evidence would let it conclude. The open items carried forward from report TDR-2026-07 are set out below; a reader should treat the committed dates as intentions rather than facts. Where the present report reports a supplier's test result, the reviewer read the certificate and did not witness the test. Two document requests remained partly unsatisfied when the report at TDR-2026-07 was delivered, and the affected conclusions are marked. Readers comparing the report at TDR-2026-07 with an earlier technical report should note that the service inventory was renamed in between.

One declinature in the present report rests on an engineering judgement the reviewer shares; the others rest on prioritisation, which is a different kind of argument. M. Quillan asked in every declined case who would revisit the decision and when, and this report records the answer even where there was none. The convention throughout report TDR-2026-07 is that an absence of evidence is reported as such rather than as evidence of absence. Grading in the report

at TDR-2026-07 is deliberately coarse, because a finer scale would imply a precision the underlying evidence does not carry. Cross-references in this report point to the evidence index rather than to a summary of the evidence.

Nothing in TDR-2026-07 is presented as verified on the strength of an interview alone. Where report TDR-2026-07 says a defect is verified, the reviewer reproduced it or read a production trace of it; everything else is recorded as a concern. Brandywine Technology Partners will not treat an open item in the present report as closed on a verbal assurance, and has said so to the engineering leadership. 10 items in this report are open, two of which the reviewer would treat as pre-closing information requests rather than post-close work. M. Quillan notes that the company answered every document request, though two answers arrived after the draft.

Report TDR-2026-07 proposes 4 metrics for the post-close monitoring pack, each already produced by an existing system. M. Quillan would add one qualitative item to the pack proposed in this report: the count of commitments closed without verifying evidence. M. Quillan interviewed 11 engineers and managers for TDR-2026-07; two declined to be quoted on one topic and that is noted where it matters. Interviewees for TDR-2026-07 were told that a stated uncertainty would be recorded as an uncertainty rather than resolved against them. The company reviewed a draft of report TDR-2026-07 for factual accuracy and its corrections are reflected without altering any conclusion.

11. Conclusions on intellectual-property hygiene

Report TDR-2026-07 concludes that the engineering claims that matter to the investment case are supportable, and two of the claims the company leads with are not. Brandywine Technology Partners concludes, on the material examined, that the engineering claims that matter to the investment case are supportable, and two of the claims the company leads with are not. In TDR-2026-07 the reviewer records the basis of each conclusion inline, so a reader can disagree with the inference without re-doing the work. TDR-2026-07 reports two findings the company disputes; the dispute is recorded next to the finding rather than resolved silently. Brandywine Technology Partners has costed remediation at the company's own loaded engineering rate rather than at a market rate.

The transaction reading of the report at TDR-2026-07 is that the engineering risk is priced in the plan rather than in the purchase agreement. Deal implications in this report are modest in aggregate and specific in detail: 2 items for the closing checklist, 3 for the plan. One limitation of report TDR-2026-07 should be stated plainly: the reviewer had no production access and relied on the company's own measurements for live latency. Report TDR-2026-07 relies on the company's incident tracker for completeness of the incident population, which the reviewer could corroborate but not prove. The third-party component inventory, licence exposure and contribution provenance of the codebase is a wide field, and TDR-2026-07 is explicit about which parts of it were reached.

4 indicators are recommended in the present report; the reviewer deliberately avoided proposing anything that needs a new pipeline. The present report would have the monitoring pack include declined items with their compensating controls, so that acceptance decisions age visibly. M. Quillan's conclusion in the report at TDR-2026-07 is that the engineering claims that matter to the investment case are supportable, and two of the claims the company leads with are not. Drawing the work together, TDR-2026-07 finds that the engineering claims that matter to the investment case are supportable, and two of the claims the company leads with are not. TDR-2026-07 contains 10 open items, and a reader should assume that number will move rather than settle.